

MERIEH AFAF HADDOU

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EDUCATION

Higher National School of Computer Science ESI (ex. INI)

Sep 2019 – Jun 2024

Software Engineer

Relevant Coursework: Software Architecture, Machine Learning, Data Mining, Database Systems, Statistics

EXPERIENCE

Fullstack Developer, Innovia

Sep 2025 – Jun 2026

Algiers, Algeria – Remote

- Architected a multi-tenant Django/PostgreSQL SaaS with 12 modules and 200+ REST APIs.
- Designed backend features including multi-tenant isolation, RBAC, and reusable search modules.
- Built reusable Vue.js components with Pinia, pagination, and lazy loading.

Python Developer, Intersec Group

Sep 2024 – Sep 2025

Paris, France – Remote

- Developed Python services processing millions of records using asyncio, batching, and AWS S3.
- Optimized a 30M-row processing pipeline with SQLite, reducing runtime by 40%.
- Built fault-tolerant data workflows with retries, caching, and automated validation.

Research Intern, LMCS / LCSi

Sep 2023 – Jun 2024

Algiers, Algeria – Hybrid

- Designed and evaluated an Optimal Transport-based drift detection system on eight datasets.
- Improved model accuracy by up to 45 percentage points (22% to 67%) through drift detection.

PROJECTS

Flood Warning IVR System (Telnyx)

Dec 2024 – May 2025

Intersec Group

- Developed an event-driven Python backend delivering real-time flood alerts.
- Integrated REST APIs, webhooks, and DTMF navigation for reliable IVR workflows.
- Reduced alert access time by 30% through backend caching optimization.

Dentix – Dental Clinic Management System

May 2026 – Jun 2026

Freelance Project

- Designed an SQLite database and optimized SQL queries, improving retrieval by 80%.
- Built a modular Python application using object-oriented design.
- Developed an Electron and Vue.js interface for clinic management.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript

Backend: FastAPI, Django, Tornado, REST APIs, API Design, Asynchronous Programming

Data Engineering: ETL Pipelines, Data Processing, Pandas, NumPy

Databases: PostgreSQL, SQLite, Relational Database Design, SQL Query Optimization

AI & Machine Learning: Scikit-learn, SciPy, Optimal Transport, Distribution Drift Detection

Cloud & Tools: Docker, AWS S3 (boto3), Git, Gerrit, Postman